Zenil Kalpesh Vadhani

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OBJECTIVE

Enthusiastic and motivated B.Tech Information Technology student with a strong foundation in programming, web development, machine learning, and computer vision. Passionate about applying technical knowledge to solve real-world problems and eager to gain hands-on experience through an internship.

EDUCATION

A. D. Patel Institute of Technology, CVM University, Anand 2023 – 2027
B.Tech Information Technology Anand, Gujarat

- **CGPA**: 8.00

INTERNSHIP

Triobyte Technologies May 2026 – Jun 2026
Summer Intern Remote

- Developed Python-based backend modules and data processing workflows in an Agile environment, contributing to scalable, modular software development for a Smart Vacation Planner system.
- Designed and implemented an AI-driven itinerary recommendation engine using ML and NLP, processing user preferences (budget, interests, duration) to generate personalized travel plans.

PROJECTS

AI-Powered Legal Document Assistant (RAG System)

- Architected a Retrieval-Augmented Generation (RAG) pipeline using LangChain and ChromaDB/FAISS vector stores for semantic search and context-aware Q&A over legal/compliance documents.
- Exposed the system via a RESTful FastAPI backend; integrated OpenAI/Gemini APIs for LLM-powered answer generation with cited source retrieval.
- **Tools**: Python, FastAPI, LangChain, ChromaDB, FAISS, OpenAI/Gemini APIs

Customer Churn Prediction System

- Engineered an end-to-end ML pipeline to predict telecom customer churn (~7,000 records); applied SMOTE for class imbalance and SHAP for model interpretability.
- Trained/benchmarked XGBoost, Random Forest, and Logistic Regression with Optuna tuning; tracked experiments via MLflow; served predictions via FastAPI.
- **Tools**: Python, XGBoost, Scikit-learn, Pandas, SHAP, Optuna, MLflow, FastAPI

Face Recognition and Attendance Management System

- Designed a real-time computer vision pipeline using OpenCV for face detection and recognition, enabling automated, contactless attendance logging.
- Trained Scikit-learn classifiers (SVM/KNN) on facial embeddings for multi-class identity recognition; automated attendance logs using NumPy.
- **Tools**: Python, OpenCV, Scikit-learn, NumPy, CSV

CERTIFICATIONS

The Ultimate Job Ready Data Science Course — CodeWithHarry Sep 2025
Certificate ID: CWH-THE-ULTIMATE-JOB-READY-DATA-SCIENCE-COURSE-QCG14KD1

Data Analytics Job Simulation — Deloitte (via Forage) Jun 2026
Completed practical tasks in data analysis and forensic technology

SKILLS

Programming Languages : C, C++, Java, Python
Frameworks & Libraries : Scikit-learn, TensorFlow, OpenCV, LangChain, FastAPI, Sentence Transformers, React.js
Core Competencies : Machine Learning, NLP, Generative AI, RAG, Vector Search, Computer Vision
Tools & Technologies : Git, AI Tools (Cursor AI, Google Antigravity)